Exact Geometric Tails and q-Binomial Extrapolation for Dyadic Values of Rvachev's Up-Function

A self-contained derivation based on the `ProveIt` development

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Abstract

Let up_n be the inverse Fourier transform of the finite product

$$\prod_{k=0}^{n} \operatorname{sinc}\left(\frac{\pi t}{2^k}\right), \qquad \operatorname{sinc} z = \frac{\sin z}{z},$$

and fix a dyadic rational $x \in [-1, 1]$. Write $s = s(x)$ for the least nonnegative integer for which $2^s x \in \mathbb{Z}$, and put $R = \lfloor s/2 \rfloor$. We prove the exact, eventually finite representation

$$\mathrm{up}_n(x) = \mathrm{up}(x) + \sum_{r=1}^{R} B_r(x) 4^{-rn}, \qquad n \geq s.$$

Thus the error is not merely asymptotic: after at most s initial levels it is a finite sum of signed geometric progressions, with ratios $4^{-1}, 4^{-2}, \ldots, 4^{-R}$. The proof combines the exact dyadic spline cells with an inverse convolution operator on a finite polynomial jet and the fact that the derivative jet of up terminates at dyadic points.

The constant term is recovered from any $R + 1$ consecutive rational samples by the single formula

$$\boxed{\mathrm{up}(x) = \frac{1}{(Q; Q)_R} \sum_{j=0}^{R} (-1)^{R-j} Q^{\binom{R-j+1}{2}} \begin{bmatrix} R \\ j \end{bmatrix}_Q \mathrm{up}_{n+j}(x), \qquad Q = \frac{1}{4}, \quad n \geq s.}$$

Here $(a; Q)_m$ is the q-Pochhammer symbol and $\begin{bmatrix} R \\ j \end{bmatrix}_Q$ is the Gaussian q-binomial coefficient. We also give an exact Vandermonde/Lagrange method for recovering every $B_r(x)$, a Romberg-style triangular implementation, stability estimates, examples, and fully rational Python verification code.

Remark 0.1 (Status and provenance). The linked `ProveIt` corpus supplies the finite-spline formula, the head–tail convolution identity, the Bernoulli–Bell reciprocal-tail coefficients, the universal q-Lagrange weights, and the global derivative formula used below [1, 2]. The exact *termination at a fixed dyadic point* is derived here by joining those ingredients through a local polynomial-jet inversion argument. This article is a mathematical proof and computational cross-check, not a claim that the new combined statement has already been formalized in Lean.

Contents

1 The result in one page

Let

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi \mathrm{i} t x}\,\mathrm{d}x \tag{1}$$

be the Fourier convention, and normalize Rvachev's function by

$$\widehat{\mathrm{up}}(t) = \prod_{k=0}^{\infty} \operatorname{sinc}\left(\frac{\pi t}{2^k}\right). \tag{2}$$

For $n \geq 0$, define the user's finite-product approximant by

$$\widehat{\mathrm{up}_n}(t) = \prod_{k=0}^{n} \operatorname{sinc}\left(\frac{\pi t}{2^k}\right). \tag{3}$$

For a dyadic x, its *dyadic depth* is

$$s(x) = \min\{s \geq 0 : 2^s x \in \mathbb{Z}\}. \tag{4}$$

Thus $s(0) = s(1) = s(-1) = 0$, while a nonintegral dyadic has a unique reduced form $x = a/2^s$ with a odd and $s = s(x)$.

The final answer is as follows.

Theorem 1.1 (Eventual exact geometric representation)**.** *Fix a dyadic $x \in [-1, 1]$, let $s = s(x)$, and set $R = \lfloor s/2 \rfloor$. There are rational numbers $B_1(x), \ldots, B_R(x)$ such that*

$$\boxed{\mathrm{up}_n(x) = \mathrm{up}(x) + \sum_{r=1}^{R} B_r(x) Q^{rn}, \qquad Q = \frac{1}{4}, \quad n \geq s.} \tag{5}$$

The coefficients have the analytic form

$$B_r(x) = (-1)^r \alpha_r 4^{-r}\, \mathrm{up}^{(2r)}(x), \tag{6}$$

where the universal rational constants α_r are determined by

$$\sum_{r=0}^{\infty} \alpha_r z^r = \prod_{\ell=1}^{\infty} \frac{1}{\operatorname{sinc}(2^{-\ell}\sqrt{z})}. \tag{7}$$

For every nonintegral dyadic of depth $s \geq 2$, the last coefficient $B_R(x)$ is nonzero, so the degree R cannot in general be reduced.

Once equation (5) is known, extraction is finite interpolation at geometric nodes. Define

$$(a; Q)_m = \prod_{k=0}^{m-1} (1 - aQ^k), \qquad \begin{bmatrix} m \\ j \end{bmatrix}_Q = \frac{(Q; Q)_m}{(Q; Q)_j (Q; Q)_{m-j}}. \tag{8}$$

Then, for any starting index $n \geq s$,

$$\boxed{\mathrm{up}(x) = \frac{1}{(Q; Q)_R} \sum_{j=0}^{R} (-1)^{R-j} Q^{\binom{R-j+1}{2}} \begin{bmatrix} R \\ j \end{bmatrix}_Q \mathrm{up}_{n+j}(x), \qquad Q = \frac{1}{4}.} \tag{9}$$

This is the requested single q-binomial/q-Pochhammer formula. It requires exactly $R + 1 = \lfloor s/2 \rfloor + 1$ rational spline values.

The rest of the article proves these statements, explains why the identity is exact rather than merely asymptotic, and gives a reproducible implementation.

2 Finite sinc products as exact dyadic splines

2.1 An indexing dictionary

It is convenient to let p_N denote the inverse transform of a prefix with exactly N factors:

$$\widehat{p}_N(t) = \prod_{k=0}^{N-1} \operatorname{sinc}\left(\frac{\pi t}{2^k}\right), \qquad N \geq 1. \tag{10}$$

The user's indexing and this factor-count indexing are related by

$$\boxed{\mathrm{up}_n = p_{n+1}.} \tag{11}$$

Keeping this one-step shift visible prevents the most common power-of-four error in the final coefficients.

For $a > 0$, let

$$b_a(x) = \frac{1}{2a}\mathbf{1}_{[-a,a]}(x). \tag{12}$$

Since $\widehat{b}_a(t) = \operatorname{sinc}(2\pi a t)$, Fourier inversion gives

$$p_N = b_{2^{-1}} * b_{2^{-2}} * \cdots * b_{2^{-N}}. \tag{13}$$

Hence p_N is a centered box spline, supported on

$$[-A_N, A_N], \qquad A_N = \sum_{j=1}^{N} 2^{-j} = 1 - 2^{-N}. \tag{14}$$

It is piecewise polynomial of degree at most $N-1$ and belongs to $C^{N-2}(\mathbb{R})$ when $N \geq 2$.

2.2 The Thue–Morse truncated-power formula

Write

$$\tau_k = (-1)^{s_2(k)}, \tag{15}$$

where $s_2(k)$ is the sum of the binary digits of k.

Proposition 2.1 (Exact finite spline). *For $N \geq 1$, away from the harmless degree-zero knot convention,*

$$p_N(y) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{k=0}^{2^N-1} \tau_k \left(y + A_N - \frac{k}{2^{N-1}}\right)_+^{N-1}. \tag{16}$$

Its knots are

$$t_{N,k} = -1 + (2k+1)2^{-N}, \qquad 0 \leq k < 2^N. \tag{17}$$

In particular, $p_N(y) \in \mathbb{Q}$ whenever $y \in \mathbb{Q}$.

Proof. Differentiate every box in equation (13). Since $\mathrm{D}b_a = (2a)^{-1}(\delta_{-a} - \delta_a)$, the distribution $\mathrm{D}^N p_N$ is a signed sum of 2^N point masses. Binary endpoint choices give the signs τ_k, the locations in equation (17), and the common multiplier $\prod_{j=1}^{N} 2^{j-1} = 2^{N(N-1)/2}$. Integrating N times from $-\infty$ yields equation (16). Rationality at rational arguments is then immediate. □

This formula is one of the central exact-arithmetic ingredients developed in the repository [2].

2.3 The exact self-similar tail

Let

$$h_N = 2^{-N}, \qquad \rho_h(y) = h^{-1}\,\mathrm{up}(y/h). \tag{18}$$

The omitted random/Fourier tail is a scaled copy of the full up-function. More precisely,

$$\boxed{\mathrm{up} = p_N * \rho_{h_N}.} \tag{19}$$

Indeed, the quotient between equation (2) and equation (10) is

$$\prod_{k=N}^{\infty} \mathrm{sinc}\left(\frac{\pi t}{2^k}\right) = \widehat{\mathrm{up}}(2^{-N}t) = \widehat{\rho_{h_N}}(t).$$

The identity is exact for every N; no limiting argument is involved.

3 Why dyadic points become cell centers

The knot grid in equation (17) has spacing 2^{1-N}. The missing-tail kernel in equation (19) has radius $h_N = 2^{-N}$, exactly half that spacing. This numerical coincidence is the geometric mechanism behind the finite answer.

Lemma 3.1 (Dyadic midpoint lemma)**.** *Let $x \in [-1, 1]$ have dyadic depth s. If $N \geq s + 1$, then the nearest knots of p_N on either side of x are*

$$x - h_N \quad \textit{and} \quad x + h_N, \tag{20}$$

with the natural exterior-cell interpretation at $x = \pm 1$. Consequently, there is a polynomial $P_{N,x}$ of degree at most $N - 1$ such that

$$p_N(x + t) = P_{N,x}(t), \qquad -h_N < t < h_N. \tag{21}$$

Proof. Since $N - 1 \geq s$,

$$M = 2^{N-1}(x + 1) \in \mathbb{Z}.$$

Therefore

$$x = -1 + 2M2^{-N}, \qquad x \pm 2^{-N} = -1 + (2M \pm 1)2^{-N}.$$

The two latter numbers are consecutive members of the odd grid equation (17). Thus x is their midpoint and p_N is a single polynomial throughout the open interval between them. □

In the user's indexing $N = n + 1$, the sufficient threshold is simply

$$\boxed{n \geq s(x).} \tag{22}$$

Earlier levels can occasionally satisfy the final formula by accident or by symmetry, but equation (22) is the uniform, explicit guarantee.

4 The reciprocal tail and its rational coefficients

4.1 Angular normalization

The inverse differential operator is cleanest in the angular Fourier variable. Put

$$\phi(\xi) = \widehat{\mathrm{up}}\left(\frac{\xi}{2\pi}\right) = \prod_{\nu=1}^{\infty} \operatorname{sinc}\left(\frac{\xi}{2^\nu}\right). \tag{23}$$

Define the reciprocal-tail function

$$\Phi(z) = \frac{1}{\phi(\sqrt{z})} = \prod_{\ell=1}^{\infty} \frac{1}{\operatorname{sinc}(2^{-\ell}\sqrt{z})} = \sum_{r=0}^{\infty} \alpha_r z^r, \qquad \alpha_0 = 1. \tag{24}$$

The nearest singularity is at $z = 4\pi^2$, so this is an ordinary analytic Taylor series near the origin. For our polynomial-operator use, only its formal coefficients matter.

4.2 Bernoulli–Bell form

The classical identity

$$\log \operatorname{sinc} w = -\sum_{k=1}^{\infty} \frac{\zeta(2k)}{k\pi^{2k}} w^{2k} \tag{25}$$

gives

$$\Phi(z) = \exp\left(\sum_{k=1}^{\infty} c_k z^k\right), \tag{26}$$

where

$$c_k = \frac{\zeta(2k)}{k\pi^{2k}(4^k-1)} = \frac{(-1)^{k+1}2^{2k-1}B_{2k}}{k(2k)!(4^k-1)} \in \mathbb{Q}. \tag{27}$$

It follows that every α_r is rational. In terms of complete exponential Bell polynomials,

$$\alpha_r = \frac{1}{r!} Y_r(1!c_1, 2!c_2, \ldots, r!c_r), \tag{28}$$

and the computationally convenient recurrence is

$$r\alpha_r = \sum_{k=1}^{r} k c_k \alpha_{r-k}. \tag{29}$$

The first values are

$$\begin{aligned} &\alpha_0 = 1, && \alpha_1 = \frac{1}{18}, && \alpha_2 = \frac{31}{16200}, \\ &\alpha_3 = \frac{2347}{42865200}, && \alpha_4 = \frac{1904369}{1311675120000}, && \alpha_5 = \frac{3309760579}{88561680752160000}. \end{aligned} \tag{30}$$

These are the same universal coefficients that occur in the repository's all-order truncation expansion [2].

Remark 4.1 (Why an asymptotic expansion is not enough). Fourier factorization gives, for each fixed truncation order M,

$$p_N = \sum_{r=0}^{M} (-1)^r \alpha_r 4^{-rN}\, \mathrm{up}^{(2r)} + O(4^{-(M+1)N})$$

in smooth norms. At a dyadic point the displayed derivatives eventually vanish, but this fact alone does *not* force the remainder to be zero. The exact result requires the local polynomial argument in the next section. That is the step which upgrades "all orders" to a finite identity.

5 Exact inversion on a finite polynomial jet

5.1 Averaging a polynomial by the tail kernel

Let Z have density up, so $Z \in [-1,1]$ and its law is even. For a polynomial P and $h > 0$, define

$$(\mathcal{C}_h P)(t) = \int_{-1}^{1} \mathrm{up}(z) P(t - hz)\,\mathrm{d}z. \tag{31}$$

This is again a polynomial of the same or lower degree.

Lemma 5.1 (Finite polynomial inverse)**.** *On the vector space of polynomials of degree at most d,*

$$\boxed{P = \Phi(-h^2\mathrm{D}^2)\,\mathcal{C}_h P = \sum_{r=0}^{\lfloor d/2 \rfloor} (-1)^r \alpha_r h^{2r} \mathrm{D}^{2r} (\mathcal{C}_h P).} \tag{32}$$

The equality is exact and purely finite.

Proof. Let $\mu_m = \int z^m \,\mathrm{up}(z)\,\mathrm{d}z$. Odd moments vanish. Taylor's formula for a polynomial gives

$$\mathcal{C}_h = \sum_{j \ge 0} \frac{\mu_{2j}}{(2j)!} h^{2j} \mathrm{D}^{2j}, \tag{33}$$

where the sum truncates on every finite-degree polynomial. Its scalar symbol is

$$M(w) = \sum_{j \ge 0} \frac{\mu_{2j}}{(2j)!} w^j = \phi(\mathrm{i}\sqrt{w}).$$

By equation (24),

$$\Phi(-w) = \frac{1}{\phi(\sqrt{-w})} = \frac{1}{\phi(\mathrm{i}\sqrt{w})} = \frac{1}{M(w)}.$$

Thus the formal power series of $\Phi(-h^2\mathrm{D}^2)$ and $\mathcal{C}_h$ are reciprocals. Because D is nilpotent on polynomials of degree at most d, their composition is exactly the identity after finitely many terms; there is no analytic convergence issue. □

5.2 The spline jet equals the averaged local polynomial jet

Lemma 5.2 (Jet transfer at a cell center)**.** *Let x have dyadic depth s, let $N \ge s+1$, set $h = h_N$, and let $P_{N,x}$ be the local polynomial from lemma 3.1. Define*

$$J_{N,x}(t) = \mathcal{C}_h P_{N,x}(t). \tag{34}$$

Then, for every $0 \le k \le N-1$,

$$J_{N,x}^{(k)}(0) = \mathrm{up}^{(k)}(x). \tag{35}$$

Equivalently,

$$J_{N,x}(t) = \sum_{k=0}^{N-1} \frac{\mathrm{up}^{(k)}(x)}{k!} t^k. \tag{36}$$

Proof. From equation (19), $\mathrm{up} = p_N * \rho_h$. For $k \le N-1$, the distributional derivative $\mathrm{D}^k p_N$ is locally integrable; point masses first appear at order N. Therefore

$$\mathrm{up}^{(k)}(x) = \int_{-h}^{h} (\mathrm{D}^k p_N)(x-y)\rho_h(y)\,\mathrm{d}y$$

$$= \int_{-1}^{1} P^{(k)}_{N,x}(-hz)\,\mathrm{up}(z)\,\mathrm{d}z = J^{(k)}_{N,x}(0).$$

The possible endpoint values at $x \pm h$ affect a set of measure zero. Both sides of equation (36) are polynomials of degree at most $N-1$ with the same derivatives at zero, so they are identical. □

Combining the preceding lemmas gives the exact local identity which drives the whole result.

Theorem 5.3 (Exact cell-center deconvolution)**.** *If x has dyadic depth s and $N \geq s+1$, then*

$$\boxed{p_N(x) = \sum_{r=0}^{\lfloor (N-1)/2 \rfloor} (-1)^r \alpha_r 4^{-rN}\,\mathrm{up}^{(2r)}(x).} \tag{37}$$

Proof. Apply lemma 5.1 to $P_{N,x}$, with degree at most $N-1$, and evaluate at zero. Since $P_{N,x}(0) = p_N(x)$, while lemma 5.2 gives $(\mathcal{C}_h P_{N,x})^{(2r)}(0) = \mathrm{up}^{(2r)}(x)$, we obtain

$$p_N(x) = \sum_{r=0}^{\lfloor (N-1)/2 \rfloor} (-1)^r \alpha_r h^{2r}\,\mathrm{up}^{(2r)}(x).$$

Now $h = 2^{-N}$, so $h^{2r} = 4^{-rN}$. □

6 The derivative jet terminates at a dyadic point

6.1 The signed global extension

Let

$$\mathcal{G}(y) = \sum_{b \geq 0} \tau_b\,\mathrm{up}(y - 2b - 1). \tag{38}$$

This sum is locally finite. On $[2b, 2b+2]$, it is simply

$$\mathcal{G}(y) = \tau_b\,\mathrm{up}(y - 2b - 1). \tag{39}$$

In particular,

$$\mathcal{G}(2m) = 0 \qquad (m \in \mathbb{Z}_{\geq 0}), \tag{40}$$

because even integers are support boundaries. On $[-1, 1]$,

$$\mathrm{up}(x) = \mathcal{G}(x+1). \tag{41}$$

The global extension satisfies the dilation equation

$$\mathcal{G}'(y) = 2\mathcal{G}(2y). \tag{42}$$

These identities are part of the arithmetic framework in [1, 5].

Iterating equation (42) gives

$$\mathcal{G}^{(m)}(y) = 2^{\binom{m+1}{2}} \mathcal{G}(2^m y), \tag{43}$$

and hence, from equation (41),

$$\boxed{\mathrm{up}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{G}(2^m(x+1))\,, \qquad -1 \leq x \leq 1.} \tag{44}$$

Lemma 6.1 (Finite dyadic jet)**.** *If x has dyadic depth s, then*

$$\mathrm{up}^{(m)}(x) = 0 \qquad \textit{for every } m > s. \tag{45}$$

Proof. For $m > s$, the number $2^m(x+1)$ is an even integer. Indeed, if $s \geq 1$, write $x = a/2^s$ with a odd; then

$$2^m(x+1) = (a + 2^s)2^{m-s},$$

and $m - s \geq 1$. The integral cases $s = 0$ are immediate as well. Now use equations (40) and (44). ☐

Remark 6.2 (The last possible mode is genuinely present)*.* Suppose $x = a/2^s$ is nonintegral and reduced. If $s = 2R$, then $2^{2R}(x+1) = a + 2^s$ is an odd integer, the center of a signed up-bump, so $\mathcal{G}$ there is ± 1. If $s = 2R+1$, then $2^{2R}(x+1) = (a+2^s)/2$ is a half-integer strictly inside a signed bump, so its $\mathcal{G}$-value is nonzero. Hence $\mathrm{up}^{(2R)}(x) \neq 0$ whenever $s \geq 2$.

7 Proof of the finite geometric formula

We can now prove the main statement in its precise coefficient form.

Theorem 7.1 (Exact finite geometric tail)**.** *Let $x \in [-1, 1]$ be dyadic, let $s = s(x)$, and put $R = \lfloor s/2 \rfloor$. For every $N \geq s + 1$,*

$$\boxed{p_N(x) = \mathrm{up}(x) + \sum_{r=1}^{R} C_r(x) 4^{-rN},} \tag{46}$$

where

$$C_r(x) = (-1)^r \alpha_r \,\mathrm{up}^{(2r)}(x). \tag{47}$$

Equivalently, in the user's indexing $q_n = \mathrm{up}_n(x) = p_{n+1}(x)$,

$$\boxed{q_n = \mathrm{up}(x) + \sum_{r=1}^{R} B_r(x) 4^{-rn}, \qquad n \geq s,} \tag{48}$$

with

$$B_r(x) = 4^{-r} C_r(x) = (-1)^r \alpha_r 4^{-r} \,\mathrm{up}^{(2r)}(x) \tag{49}$$

$$= (-1)^r \alpha_r 2^{r(2r-1)} \mathcal{G}\Big(2^{2r}(x+1)\Big). \tag{50}$$

All coefficients are rational. If $s \geq 2$, then $B_R(x) \neq 0$.

Proof. The cell-center identity equation (37) initially has terms up to $\lfloor (N-1)/2 \rfloor$. By lemma 6.1, every derivative of order larger than s is zero, so only $2r \leq s$, or $r \leq R$, remains. This proves equation (46). Substitute $N = n + 1$ and absorb one factor 4^{-r} into the coefficient to obtain equation (48). Equation (50) follows from equation (44), because

$$4^{-r} 2^{\binom{2r+1}{2}} = 2^{-2r} 2^{r(2r+1)} = 2^{r(2r-1)}.$$

The rationality of α_r follows from equation (27). Values of $\mathcal{G}$ at dyadic arguments are rational by the standard arithmetic theory of the Fabius/up function [5, 1]; alternatively, rationality of every coefficient follows directly by solving the rational Vandermonde system in section 9. The nonvanishing of the last coefficient follows from the preceding remark and $\alpha_R > 0$. ☐

Thus the error sequence is a finite linear combination of signed geometric progressions:

$$q_n - \mathrm{up}(x) = B_1(x)(1/4)^n + B_2(x)(1/16)^n + \cdots + B_R(x)(4^{-R})^n. \tag{51}$$

Each progression decreases in absolute value as n increases. The number of nonconstant modes is bounded by, and for $s \geq 2$ reaches, the explicit order $R = \lfloor s/2 \rfloor$.

Corollary 7.2 (An exact recurrence)**.** *Let E be the forward shift $(Eq)_n = q_{n+1}$. On the tail $n \geq s$,*

$$\prod_{r=1}^{R}(E - Q^r)(q_n - \mathrm{up}(x)) = 0, \qquad Q = \frac{1}{4}, \tag{52}$$

and therefore

$$(E-1)\prod_{r=1}^{R}(E - Q^r)q_n = 0. \tag{53}$$

Corollary 7.3 (Rational tail generating function)**.** *For $n \geq s$,*

$$\sum_{m=0}^{\infty}(q_{s+m} - \mathrm{up}(x))z^m = \sum_{r=1}^{R}\frac{B_r(x)Q^{rs}}{1 - Q^r z}. \tag{54}$$

8 Exact extraction by q-binomial interpolation

8.1 The interpolation problem

Fix any starting index $n \geq s$, set $Q = 1/4$, and rewrite equation (48) as

$$q_{n+j} = L + \sum_{r=1}^{R} D_r (Q^j)^r, \qquad j = 0, 1, \ldots, R, \tag{55}$$

where

$$L = \mathrm{up}(x), \qquad D_r = B_r(x)Q^{rn}. \tag{56}$$

Thus $q_{n+j} = g_n(Q^j)$ for a polynomial

$$g_n(z) = L + D_1 z + \cdots + D_R z^R. \tag{57}$$

The desired limit is simply $L = g_n(0)$.

8.2 Closed q-Pochhammer weights

For $m \geq 0$, recall

$$(a;Q)_m = \prod_{k=0}^{m-1}(1 - aQ^k), \qquad (Q;Q)_m = \prod_{k=1}^{m}(1 - Q^k), \tag{58}$$

and

$$\begin{bmatrix} m \\ j \end{bmatrix}_Q = \frac{(Q;Q)_m}{(Q;Q)_j (Q;Q)_{m-j}}. \tag{59}$$

Theorem 8.1 (Universal q-Lagrange weights)**.** *For* $0 < Q < 1$*, define*

$$w_j^{(R)} = \frac{(-1)^{R-j} Q^{\binom{R-j+1}{2}}}{(Q;Q)_j (Q;Q)_{R-j}} = \frac{(-1)^{R-j} Q^{\binom{R-j+1}{2}}}{(Q;Q)_R} \begin{bmatrix} R \\ j \end{bmatrix}_Q. \tag{60}$$

Then

$$\sum_{j=0}^{R} w_j^{(R)} = 1, \qquad \sum_{j=0}^{R} w_j^{(R)} Q^{jr} = 0 \quad (1 \le r \le R). \tag{61}$$

Consequently, for the dyadic spline sequence,

$$\boxed{\operatorname{up}(x) = \sum_{j=0}^{R} w_j^{(R)} \operatorname{up}_{n+j}(x), \qquad n \ge s.} \tag{62}$$

Equivalently,

$$\boxed{\operatorname{up}(x) = \frac{1}{(Q;Q)_R} \sum_{j=0}^{R} (-1)^{R-j} Q^{\binom{R-j+1}{2}} \begin{bmatrix} R \\ j \end{bmatrix}_Q \operatorname{up}_{n+j}(x), \quad Q = \frac{1}{4}, \ n \ge s.} \tag{63}$$

Proof. The Lagrange coefficient for evaluating at zero from the nodes $1, Q, \dots, Q^R$ is

$$\ell_j(0) = \prod_{\substack{0 \le k \le R \\ k \ne j}} \frac{-Q^k}{Q^j - Q^k}. \tag{64}$$

Separate the factors with $k < j$ and $k > j$. The signs, powers of Q, and products $1 - Q^m$ simplify to

$$\ell_j(0) = \frac{(-1)^{R-j} Q^{(R-j)(R-j+1)/2}}{(Q;Q)_j (Q;Q)_{R-j}},$$

which is equation (60). Lagrange interpolation is exact for every polynomial of degree at most R; applying it to $1, z, \dots, z^R$ yields equation (61). Applying it to g_n in equation (57) yields equation (62), and the Gaussian form gives equation (63). □

The first three filters, with $Q = 1/4$, are

$$R = 1: \quad \operatorname{up}(x) = \frac{4q_{n+1} - q_n}{3}, \tag{65}$$

$$R = 2: \quad \operatorname{up}(x) = \frac{q_n - 20q_{n+1} + 64q_{n+2}}{45}, \tag{66}$$

$$R = 3: \quad \operatorname{up}(x) = \frac{-q_n + 84q_{n+1} - 1344q_{n+2} + 4096q_{n+3}}{2835}. \tag{67}$$

8.3 A triangular Richardson table

The weighted sum can be evaluated without constructing all weights at once.

Algorithm 8.2 (Exact q-Richardson table)**.** Start with

$$T_{0,n} = q_n. \tag{68}$$

For $m = 1, 2, \dots, R$, form

$$\boxed{T_{m,n} = \frac{T_{m-1,n+1} - Q^m T_{m-1,n}}{1 - Q^m}.} \tag{69}$$

Then

$$T_{R,n} = \operatorname{up}(x) \qquad (n \ge s). \tag{70}$$

At stage m, the new row removes the mode Q^{mn}. This is the Romberg-style realization of the polynomial

$$W_R(E) = \prod_{m=1}^{R} \frac{E - Q^m}{1 - Q^m}. \tag{71}$$

The formulas and their moment identities are also developed in the repository's q-Richardson layer [2].

9 Recovering every coefficient from the same samples

The extraction formula is the first row of an ordinary rational Vandermonde problem. From equation (55),

$$\begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & Q & Q^2 & \cdots & Q^R \\ 1 & Q^2 & Q^4 & \cdots & Q^{2R} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & Q^R & Q^{2R} & \cdots & Q^{R^2} \end{pmatrix} \begin{pmatrix} L \\ D_1 \\ D_2 \\ \vdots \\ D_R \end{pmatrix} = \begin{pmatrix} q_n \\ q_{n+1} \\ q_{n+2} \\ \vdots \\ q_{n+R} \end{pmatrix}. \tag{72}$$

Its determinant is

$$\prod_{0 \le i < j \le R} (Q^j - Q^i) \neq 0, \tag{73}$$

so the solution is unique. Since the matrix and data are rational, every L, D_r, and hence every $B_r = Q^{-rn} D_r$, is rational.

An explicit all-coefficient interpolation formula is

$$g_n(z) = \sum_{j=0}^{R} q_{n+j} \prod_{\substack{0 \le k \le R \\ k \neq j}} \frac{z - Q^k}{Q^j - Q^k}. \tag{74}$$

Then

$$\boxed{\mathrm{up}(x) = g_n(0), \qquad B_r(x) = Q^{-rn}[z^r]g_n(z) \quad (1 \le r \le R).} \tag{75}$$

Thus the same $R + 1$ values determine both the true rational limit and every geometric correction coefficient.

10 A practical exact-arithmetic method

Algorithm 10.1 (Limit extraction at a dyadic point)**.** Given a dyadic $x \in [-1, 1]$:

1. Reduce x and compute its dyadic depth $s = \min\{j : 2^j x \in \mathbb{Z}\}$.
2. Set $R = \lfloor s/2 \rfloor$, $Q = 1/4$, and choose any $n \ge s$. The minimal choice is $n = s$.
3. Evaluate the $R+1$ rational spline values $q_n, q_{n+1}, \ldots, q_{n+R}$, for example by equation (16).
4. Apply the single formula equation (63), or equivalently construct the triangular table equation (69). The result is exactly $\mathrm{up}(x)$.
5. If the correction terms are wanted, solve equation (72) or use equation (75).

6. As an exact diagnostic, repeat the extraction from the next overlapping block. Every block beginning at $n \geq s$ must return the same rational number.

No estimate of a small remainder is needed. With rational arithmetic, equality is literal numerator/denominator equality.

10.1 Stability for inexact inputs

If the input samples are rounded rather than exact, the absolute condition number of the weighted sum is

$$\Lambda_R(Q) = \sum_{j=0}^{R} \left| w_j^{(R)} \right| = \frac{(-Q;Q)_R}{(Q;Q)_R}. \tag{76}$$

For $Q = 1/4$, these values remain below the convergent limit

$$\Lambda_\infty(1/4) = \frac{(-1/4;1/4)_\infty}{(1/4;1/4)_\infty} \approx 1.9692603537. \tag{77}$$

Thus even high-order dyadic filters have modest absolute amplification. Exact rational arithmetic, of course, eliminates roundoff entirely.

10.2 Over-ordering

If one uses any order $M \geq R$, the same weight formula with M in place of R also returns $\mathrm{up}(x)$ exactly, because it annihilates every actual mode. The minimal order R uses the fewest samples and normally gives the smallest arithmetic workload.

11 Worked exact examples

11.1 Depth two: one geometric mode

For $x = 1/4$, $s = 2$ and $R = 1$. Exact spline evaluation gives

$$\mathrm{up}_n\left(\frac{1}{4}\right) = \frac{67}{72} + \frac{1}{9}4^{-n}, \qquad n \geq 2. \tag{78}$$

For example,

$$q_2 = \frac{15}{16}, \qquad q_3 = \frac{179}{192},$$

and equation (65) gives

$$\mathrm{up}\left(\frac{1}{4}\right) = \frac{4(179/192) - 15/16}{3} = \frac{67}{72}. \tag{79}$$

The fitted coefficient and $\alpha_1 = 1/18$ imply $\mathrm{up}''(1/4) = -8$, consistent with equation (44).

Similarly,

$$\mathrm{up}_n\left(\frac{3}{4}\right) = \frac{5}{72} - \frac{1}{9}4^{-n}, \qquad n \geq 2, \tag{80}$$

so the same filter returns $\mathrm{up}(3/4) = 5/72$.

11.2 Depth three: still one even-derivative mode

For $x = 1/8$, $s = 3$ but $R = \lfloor 3/2 \rfloor = 1$, because the correction operator uses only even derivatives. The exact identity is

$$\mathrm{up}_n\left(\frac{1}{8}\right) = \frac{287}{288} + \frac{1}{18}4^{-n}, \qquad n \geq 3. \tag{81}$$

Thus two consecutive values again suffice.

11.3 Depth four: two geometric modes

For $x = 1/16$, $s = 4$ and $R = 2$. The three minimal samples are

$$q_4 = \frac{24575}{24576}, \qquad q_5 = \frac{1965959}{1966080}, \qquad q_6 = \frac{94365509}{94371840}. \tag{82}$$

The order-two filter gives

$$\mathrm{up}\left(\frac{1}{16}\right) = \frac{q_4 - 20q_5 + 64q_6}{45} = \frac{2073457}{2073600}. \tag{83}$$

Solving the full system yields

$$\boxed{\mathrm{up}_n\left(\frac{1}{16}\right) = \frac{2073457}{2073600} + \frac{5}{648}4^{-n} - \frac{248}{2025}16^{-n}, \qquad n \geq 4.} \tag{84}$$

The corresponding derivatives inferred from equation (49) are

$$\mathrm{up}''\left(\frac{1}{16}\right) = -\frac{5}{9}, \qquad \mathrm{up}^{(4)}\left(\frac{1}{16}\right) = -1024. \tag{85}$$

11.4 A table of exact models

x	s	R	$\mathrm{up}(x)$	B_1	B_2
0	0	0	1	–	–
1/2	1	0	1/2	–	–
1/4	2	1	67/72	1/9	–
3/4	2	1	5/72	−1/9	–
1/8	3	1	287/288	1/18	–
1/16	4	2	2073457/2073600	5/648	−248/2025
3/16	4	2	2026943/2073600	67/648	248/2025
5/16	4	2	1767743/2073600	67/648	248/2025
1/32	5	2	33177581/33177600	1/2592	−124/2025

In each row the model is $\mathrm{up}_n(x) = \mathrm{up}(x) + B_1 4^{-n} + B_2 16^{-n}$ with absent terms omitted and with validity from $n \geq s$.

12 Computational verification

The archive accompanying this article contains

`exact_extrapolation.py`

and its generated report

`verification_output.txt`.

The program uses only the Python standard library. Every quantity is a `fractions.Fraction`; no floating-point value enters any identity check. It implements:

- the Thue–Morse truncated-power formula for $p_N(x)$;
- q-Pochhammer symbols, Gaussian binomials, and the weights equation (60);
- the triangular table equation (69);
- exact Gauss–Jordan solution of equation (72);
- Bernoulli numbers, the cumulants c_k, and the recurrence for α_r;
- overlapping-block and out-of-sample checks of every recovered model.

The included run verifies all signed dyadic points of depth at most six, including endpoints: 129 points in total. At each point it checks multiple overlapping extraction blocks, the full fitted formula at later levels, and the annihilating moment identities. All checks pass exactly. These experiments are evidence against indexing or sign mistakes; the proof itself is contained in lemmas 3.1, 5.1, 5.2 and 6.1 and theorems 7.1 and 8.1.

To reproduce the report from the archive, run

```
python exact_extrapolation.py --max-depth 6 \
  --output verification_output.txt
```

Larger depths may be requested, at an exponentially growing cost for the straightforward truncated-power evaluator.

13 Further consequences and extensions

13.1 The universal part and the dyadic part

Two logically distinct structures meet in the result:

(i) The reciprocal tail and the q-filter are universal for geometric sinc products. For base $b > 1$, the natural ratio is $Q = b^{-2}$, and the same weights equation (60) annihilate the modes $Q^n, \dots, Q^{Rn}$.

(ii) Exact finite termination here depends on special dyadic geometry: the point becomes the center of a uniform spline cell, and the global derivative orbit reaches an even integer after finitely many doublings.

Thus q-Richardson acceleration exists much more generally, while exact finite recovery is an arithmetic superconvergence phenomenon.

13.2 Nonconsecutive levels

Consecutive levels are not essential. Given distinct levels $n_0, \ldots, n_R \geq s$, interpolate at the nodes $Q^{n_0}, \ldots, Q^{n_R}$ and evaluate at zero:

$$\mathrm{up}(x) = \sum_{j=0}^{R} q_{n_j} \prod_{\substack{0 \leq k \leq R \\ k \neq j}} \frac{-Q^{n_k}}{Q^{n_j} - Q^{n_k}}. \tag{86}$$

The consecutive choice is preferable because the common factor Q^n cancels and produces the compact q-Pochhammer expression.

13.3 Exact certification from data

When x is known, the depth supplies a proof-level order bound, so there is no model-selection problem. Nevertheless, the following checks are useful in software:

- $T_{R,n} = T_{R,n+1}$ for overlapping blocks;
- the recovered coefficients reproduce later q_k exactly;
- the moment sums in equation (61) vanish exactly;
- the highest fitted coefficient is nonzero for reduced depth $s \geq 2$.

Failure of any check signals a wrong index convention, an insufficient start level, or an error in the spline evaluator.

14 Conclusion

At a fixed dyadic rational, the finite sinc-product splines exhibit a stronger phenomenon than ordinary convergence. From level $n = s(x)$ onward, the point is centered in one polynomial cell whose width matches the support of the self-similar omitted tail. The tail convolution can therefore be inverted exactly on a finite polynomial jet. The global doubling equation then kills all derivatives above the dyadic depth. What appears globally as an infinite all-order expansion collapses pointwise to

$$\mathrm{up}_n(x) = \mathrm{up}(x) + \sum_{r=1}^{\lfloor s(x)/2 \rfloor} B_r(x) 4^{-rn}.$$

The constant term of this finite geometric polynomial is evaluation at zero from the nodes $1, Q, \ldots, Q^R$. Lagrange interpolation on that geometric grid is exactly the q-binomial filter

$$\mathrm{up}(x) = \frac{1}{(Q;Q)_R} \sum_{j=0}^{R} (-1)^{R-j} Q^{\binom{R-j+1}{2}} \begin{bmatrix} R \\ j \end{bmatrix}_Q \mathrm{up}_{n+j}(x), \qquad Q = \frac{1}{4}.$$

It uses finitely many rational inputs and returns the true rational value with no limiting process and no remainder estimate.

A Compact derivation of the weight formula

For reference, start from

$$w_j = \prod_{k \neq j} \frac{-Q^k}{Q^j - Q^k}.$$

For $k < j$, write

$$Q^j - Q^k = -Q^k(1 - Q^{j-k});$$

for $k > j$, write

$$Q^j - Q^k = Q^j(1 - Q^{k-j}).$$

There are j signs from the first group. The denominator products are $(Q;Q)_j$ and $(Q;Q)_{R-j}$. The remaining power of Q is

$$\sum_{k \neq j} k - \left(\sum_{k<j} k + j(R-j)\right) = \frac{(R-j)(R-j+1)}{2}.$$

Combining signs and powers gives equation (60).

B Normalization summary

Object	Definition
Full up-function	$\widehat{\mathrm{up}}(t) = \prod_{k\geq 0} \mathrm{sinc}(\pi t/2^k)$ under the two-pi convention.
User's approximant	$\widehat{\mathrm{up}_n}(t) = \prod_{k=0}^{n} \mathrm{sinc}(\pi t/2^k)$.
Factor-count spline	$p_N = \mathrm{up}_{N-1}$, containing exactly N factors.
Tail radius	$h_N = 2^{-N}$, with $\mathrm{up} = p_N * \rho_{h_N}$.
Geometric ratio	$Q = 1/4$.
Dyadic depth	$s(x) = \min\{s \geq 0 : 2^s x \in \mathbb{Z}\}$.
Number of modes	$R = \lfloor s(x)/2 \rfloor$.
Guaranteed user level	$n \geq s(x)$.

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